

VALENCE

Version Report & Technical Documentation

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Technology: C++17 | Qt6 | Windows | CMake

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1. Project Overview

Valence is a minimal, high-performance code editor built entirely from scratch in C++17 and Qt6. It targets Windows as a native desktop application, compiled to a standalone .exe with zero external runtime dependencies beyond the Qt framework. The editor uses a custom QPainter-based rendering pipeline (no QTextEdit) with viewport culling for 60fps performance, a line-array document model, command-pattern undo/redo with temporal grouping, and a single-pass O(n) C++ syntax highlighter.

Language	C++17
Framework	Qt6 (Widgets)
Platform	Windows (x64)
Build System	CMake 3.16+
Source Files	20
Lines of Code	~2,500
External Dependencies	None (Qt only)

2. Architecture

The codebase follows a strict layered architecture. Each layer depends only on layers below it.

Theme	src/theme/	theme.h	Color palette, font configuration, UI constants
Core	src/core/	TextBuffer, UndoManager, Selection	Document model, edit operations, cursor state
Editor	src/editor/	EditorWidget, CppHighlighter	Custom rendering, syntax tokenization, input handling
UI	src/ui/	MainWindow, TabWidget, FileExplorer, TerminalWidget, StatusBar	Application shell, layout, menus, terminal

3. Version History

3.1. V1.0 — Foundation Release

Release Date: 2026-04-28

Status: Basic editing functional. Shortcuts non-operational. UI functional but rough.

Summary

Initial implementation of the Valence editor. Built the complete multi-layer architecture from scratch: Core data structures (TextBuffer, UndoManager, Selection), a custom QPainter-based editor widget with viewport culling and C++ syntax highlighting, and the full UI shell (tabbed interface, sidebar file explorer, embedded PowerShell terminal, and status bar). All 20 source files organized under src/ with a CMake build system.

Files Affected

src/theme/theme.h	NEW	Centralized color palette and font configuration
src/core/selection.h	NEW	Position and Selection structures

src/core/text_buffer.h	NEW	TextBuffer header — line-array document model
src/core/text_buffer.cpp	NEW	TextBuffer implementation — insert, delete, split, merge, file I/O
src/core/undo_manager.h	NEW	UndoManager header — command-pattern undo/redo
src/core/undo_manager.cpp	NEW	UndoManager implementation — 400ms temporal grouping
src/editor/cpp_highlighter.h	NEW	CppHighlighter header — single-pass O(n) tokenizer
src/editor/cpp_highlighter.cpp	NEW	CppHighlighter implementation — keyword/type/string/comment detection
src/editor/editor_widget.h	NEW	EditorWidget header — custom QPainter editor
src/editor/editor_widget.cpp	NEW	EditorWidget implementation — rendering, input, selection, clipboard
src/ui/main_window.h	NEW	MainWindow header
src/ui/main_window.cpp	NEW	MainWindow implementation — layout, menus, signal wiring
src/ui/tab_widget.h	NEW	TabWidget header
src/ui/tab_widget.cpp	NEW	TabWidget implementation — styled tabs with close buttons
src/ui/file_explorer.h	NEW	FileExplorer header
src/ui/file_explorer.cpp	NEW	FileExplorer implementation — QTreeView + QFileSystemModel
src/ui/terminal_widget.h	NEW	TerminalWidget header
src/ui/terminal_widget.cpp	NEW	TerminalWidget implementation — persistent PowerShell session
src/ui/status_bar.h	NEW	StatusBar header
src/ui/status_bar.cpp	NEW	StatusBar implementation — file, cursor, language, encoding
src/main.cpp	NEW	Application entry point
CMakeLists.txt	MODIFIED	Reorganized for src/ structure, renamed project to Valence

Design Decisions

- Editor: 100% custom QPainter — no QTextEdit. Viewport culling renders only visible lines.
- Cursor: Neon teal #00ff9c with glow effect, 500ms blink timer.
- Background hierarchy: Editor #0b0f14 → Sidebar #0a0d12 → Terminal #070a0e.
- Tabs: Accent underline style (no heavy boxes), modified indicator (●).
- Sidebar: 2px left accent bar on selected file.
- Terminal: Persistent PowerShell session with 1px top border separator.
- Tab key: 4 spaces (soft tabs). Font: JetBrains Mono with Consolas fallback.

3.2. V1.1 — Stability & UI Overhaul

Release Date: 2026-04-29

Status: Shortcuts fully operational. File management functional. UI refined to premium tier.

Summary

Critical crash fixes for keyboard shortcuts (infinite recursion from synthetic QKeyEvents), complete rewrite of key handling with clean Ctrl/non-Ctrl separation, Tab key interception, auto-bracket closing. File Explorer overhauled with action toolbar (New File, New Folder, Refresh, Collapse All). Premium Apple-inspired theme refresh with warmer backgrounds, softer borders, and refined menu styling.

Files Affected

src/editor/editor_widget.h	MODIFIED	Made undo/redo/clipboard methods public; added event() override for Tab interception
src/editor/editor_widget.cpp	MODIFIED	Rewrote keyPressEvent (Ctrl/non-Ctrl split); fixed handleTab undo position; fixed handleDelete merge duplication; added auto-bracket closing; added event() override
src/ui/main_window.cpp	MODIFIED	Removed synthetic QKeyEvent shortcuts (crash fix); edit menu calls editor methods directly; refined stylesheet with rounded menus, tooltip styling
src/ui/file_explorer.h	MODIFIED	Added toolbar buttons, project label, new file/folder signals
src/ui/file_explorer.cpp	MODIFIED	Complete overhaul — action toolbar, file/folder creation dialogs, project name display
src/ui/tab_widget.cpp	MODIFIED	Refined tab styling — softer proportions, smaller font, drawBase disabled
src/ui/terminal_widget.cpp	MODIFIED	Softer scrollbars, focus accent on input, refined padding
src/theme/theme.h	MODIFIED	Warmer backgrounds (#16161c base), Material-inspired syntax palette, subtler borders

Bug Fixes

- CRITICAL: Menu QAction shortcuts caused infinite recursion via synthetic QKeyEvents → stack overflow crash on Ctrl+V/Z/A/C/X.
- Letters a, c, v, x, z, y, s could not be typed — switch cases broke even when Ctrl was not held.
- Tab key intercepted by Qt's focus navigation system — overrode event() to capture Tab/Backtab.
- handleTab recorded undo position AFTER cursor moved → undo accessed invalid buffer positions.
- handleDelete manually appended line AND called mergeLines → duplicated line content.

3.3. V1.2 — Build & Run Integration

Release Date: 2026-04-29

Status: Compile-and-run pipeline operational. Syntax highlighting vivid and colorful.

Summary

Added Run menu with Build & Run (F5) and Build Only (Ctrl+Shift+B). Auto-saves the current file, cd's to its directory, compiles with g++, and executes the resulting .exe — all piped through the integrated terminal. Syntax colors boosted to be more vibrant and distinct.

Files Affected

src/ui/main_window.h	MODIFIED	Added runCurrentFile() and buildCurrentFile() slots
src/ui/main_window.cpp	MODIFIED	Added Run menu (F5, Ctrl+Shift+B); implemented build/run methods with auto-save, cd, g++, and exe execution; added QFileInfo include
src/ui/terminal_widget.h	MODIFIED	Added public runCommand() and clearOutput() methods
src/ui/terminal_widget.cpp	MODIFIED	Implemented runCommand() and clearOutput() for programmatic terminal control
src/theme/theme.h	MODIFIED	Boosted syntax vibrancy — sky blue keywords, bright green functions, warm orange numbers, more visible punctuation

Design Decisions

- Build & Run (F5): auto-save → cd to file dir → g++ compile → execute .exe
- Build Only (Ctrl+Shift+B): compile without execution
- PowerShell chaining: g++ file.cpp -o file.exe ; if (\$?) { .\file.exe }
- Terminal auto-shows and clears on each run for clean output

4. File Manifest

Complete listing of all source files in the current release.

src/main.cpp	Application entry point
src/theme/theme.h	Color palette, font config, UI constants
src/core/selection.h	Position and Selection structures
src/core/text_buffer.h	TextBuffer class declaration
src/core/text_buffer.cpp	Line-array document model implementation
src/core/undo_manager.h	UndoManager class declaration
src/core/undo_manager.cpp	Command-pattern undo/redo with temporal grouping
src/editor/cpp_highlighter.h	CppHighlighter class declaration
src/editor/cpp_highlighter.cpp	Single-pass O(n) C++ syntax tokenizer
src/editor/editor_widget.h	EditorWidget class declaration
src/editor/editor_widget.cpp	Custom QPainter rendering, input handling, clipboard
src/ui/main_window.h	MainWindow class declaration
src/ui/main_window.cpp	Top-level window, menus, shortcuts, file operations
src/ui/tab_widget.h	TabWidget class declaration
src/ui/tab_widget.cpp	Multi-tab management with styled tab bar
src/ui/file_explorer.h	FileExplorer class declaration
src/ui/file_explorer.cpp	Sidebar file tree with action toolbar
src/ui/terminal_widget.h	TerminalWidget class declaration
src/ui/terminal_widget.cpp	Embedded PowerShell terminal
src/ui/status_bar.h	StatusBar class declaration
src/ui/status_bar.cpp	File name, cursor position, language, encoding display
CMakeLists.txt	Build configuration

5. Design Decisions

- **Custom Rendering:** QPainter-based rendering with viewport culling instead of QTextEdit. Only visible lines are drawn, ensuring 60fps even on large files.
- **Line-Array Model:** Document stored as `std::vector<std::string>`. O(1) line access, O(n) for line-internal edits. Optimal for files under 100K lines.
- **Temporal Undo Grouping:** Consecutive edits within 400ms at adjacent positions are grouped into a single undo action, matching user expectations.
- **No QTextEdit:** Complete input handling in keyPressEvent with clean Ctrl/non-Ctrl separation. Avoids Qt's text widget overhead and behavior quirks.
- **Soft Tabs:** Tab key inserts 4 spaces. No tab character support in V1.
- **PowerShell Terminal:** Persistent PowerShell process via QProcess. Commands piped via stdin. ANSI escape sequences stripped for clean output.
- **Theme System:** All colors and fonts defined as inline constants in theme.h. Single source of truth for the entire UI.

— End of Report —